PROJECT REPORT
ON
"Miniaturized Jug-Shaped CPW-Fed UWB Antenna for Wireless Communication"

In partial fulfilment of the requirements for the Award of the degree

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

In the subject of

PROJECT WORK [21ECP76]

Submitted by:

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CHIKKAMAGALURU – 577102

2024-2025

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(2024-2025)

DEPARTMENT OF ELECTRONICS AND COMMUNICTION ENGINEERING

CERTIFICATE

This is to certify that the Main-project work report (21ECP76) work entitled **"Miniaturized Jug-Shaped CPW-Fed UWB Antenna for Wireless Communication"** is a Bonafide work carried out by **PRAGNA J (4AI21EC055), SINCHANA K S (4AI21EC084), SINCHANA KAKADE S (4AI21EC085), SOWMYA C G (4AI21EC089)** students of 7th Semester B.E., in partial fulfilment for the Main-Project Work of 7th semester Bachelor of Engineering in Electronics and Communication Engineering of the Visvesvaraya Technological University, Belagavi during the academic year 2024-2025. It is certified that all corrections and suggestions indicated for Internal Assessment have been incorporated in the report deposited in the department library. The Main-project report has been approved, as it satisfies the academic requirements in respect of Main-Project Work prescribed for the said degree.

03.01.25

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EXTERNAL VIVA

Name of the examiner | **Signature with date**

DECLARATION

We, **PRAGNA J (4AI21EC055), SINCHANA K.S (4AI21EC084), SINCHANA KAKADE S (4AI21EC085), SOWMYA C.G (4AI21EC089)** students of 7th semester B.E., Electronics and Communication Engineering, Adichunchanagiri Institute of Technology, Chikkmagaluru, hereby declare that Main-Project Work entitled **"Miniaturized Jug-Shaped CPW-Fed UWB Antenna for Wireless Communication"** has been carried out by us under the guidance of **Mr. RAJAPPA H S** Assistant Professor Dept. of Electronics and Communication Engineering, AIT, Chikkmagaluru, in the partial fulfilment for the Project Work of 7th semester Bachelor of Engineering in Electronics and Communication Engineering of the Visvesvaraya Technological University, Belagavi during the academic year 2024-2025. We also declare that, to the best of our knowledge and belief, the matter embodied in this Project Report has not been submitted previously by us for the award of any Degree or Diploma to any other university.

Date: 06/01/25
Place: Chikkmagaluru.

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ABSTRACT

This project presents the design, analysis, and evaluation of a miniaturized coplanar waveguide (CPW)-fed ultra-wideband (UWB) Jug-Shaped Antenna tailored for wireless applications. It addresses the challenges of miniaturization, wide bandwidth, and efficient performance. The operating frequency refers to the range where the system demonstrates optimal performance, typically defined by, ensuring efficient power transfer and good impedance matching. This project analyses the return loss characteristics of a CPW-fed UWB antenna operating across the frequency bands 2.6383–3.6896 GHz, 4.6596–5.0724 GHz, and 6.5805–9.4254 GHz. The antenna demonstrates return loss values below -10 dB, indicating excellent impedance matching and efficient power transfer. The CPW-fed design enhances wideband impedance matching, enabling effective integration into compact systems. This performance makes the antenna suitable for diverse applications, including Wi-Fi (2.4 GHz and 5 GHz), 5G mid-band communication, satellite communication (C-band and X-band), military radar, and remote sensing. The low return loss underscores its potential for high-performance multi-band wireless communication and sensing systems. The Jug-Shaped Antenna combines miniaturization, wideband operation, and efficient performance, making it a strong candidate for next-generation wireless communication systems.

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Chapter 01

INTRODUCTION

1.1 Overview

Ultra-wideband (UWB) technology is a telecommunications technology that enables high-speed bandwidth connections with low energy consumption in wireless communication networks. The UWB was designed within commercial radar applications in mind at first. Nowadays, UWB technology has two primary applications: wireless personal area networks (PANs) and consumer electronics. After its initial success in the mid-2000s, UWB radio evolved into a technology within unique smart structures such as radar, wireless communications, and medical applications.

Up until 2001, UWB was used in military applications only. The Federal Communications Commission (FCC) permitted the general public's use of UWB capacity for commercial purposes after 2002. Furthermore, the FCC permitted the use of the UWB spectrum, which in the United States lies between 3.1 and 10.6 GHz. Short-range transmission of UWB is due to its low spectral density. However, high gain antennas with relatively constant radiation characteristics are necessary for this function. Planar antennas, particularly monopoles, are used in UWB equipment due to their compact size, low profile, low cost and ultra-wide impedance bandwidth. Additionally, positioning these antennas near metallic surfaces may cause a considerable impedance mismatch. When transmitting restricted frequency signals, low-profile antennas have low gain and poor directivity.

The UWB technology advantages are high data rates, slight interference, reliability, cost-effectiveness, and low complexity. As the pulse time is a fraction of a millisecond, this method has one major drawback, including the need for precise receptor time synchronization. Radar, medical imaging, and military communications are just a few of the use areas for this technology. UWB antennas with band rejection capabilities have recently been developed to reduce interference in wireless applications with limited bandwidth. Various slots can be inserted into the patch, feed, and ground plane to accomplish this approach. A reduced antenna size to boost the bandwidth using a slot has been reported. New antenna structures are being sought to suit the strict requirements of smart gadgets compatible with current wireless communications.

1.2 Literature Survey

A compact CPW-fed ultra-wideband (UWB) antenna with dual-notched bands at 4–5.78 GHz and 6.83–8.22 GHz has been designed, featuring dimensions of just 20 × 23 mm². The antenna employs an octagonal patch on an FR-4 substrate, achieving a wide impedance matching range of 7.33 GHz (3.2–10.5 GHz). The notched bands are created using split-ring circular slots, allowing for independent control over the notching frequencies. With a high efficiency of over 90% and a stable gain peaking at 4.8 dB, this antenna is well-suited for integration into small devices. Additionally, its design facilitates easy mass production, making it an excellent candidate for modern compact communication systems [1].

Limitations:

- The dual-notch bands limit continuous coverage, which may not be ideal for applications needing uninterrupted UWB operation.
- Focuses primarily on size, gain, and efficiency; lacks detailed analysis of advanced parameters like group delay and CCL.
- The independent control of notch bands adds complexity to both design and validation processes.

A SAPM antenna with a bandwidth ranging from 3.05 GHz to 11.9 GHz (8.85 GHz) is designed for enhanced performance. It incorporates a frequency-selective surface (FSS) reflector to improve gain. The radiating element features a six-cylinder intersection, providing an artistic shape. The FSS reflector consists of a 10 × 10 array with unit cell dimensions of 6 mm × 6 mm, achieving a stop-band transmission coefficient of less than −10 dB and a linear reflection phase across the operating band. The gain of the antenna is significantly enhanced with the FSS reflector, increasing from 1.65 dB to 7.87 dB in the lower band and from 6.3 dB to 9.68 dB in the upper band. The compact antenna, with dimensions of 61 × 61 × 1.6 mm, is suitable for ultra-wideband (UWB) systems and ground-penetrating radar (GPR) applications, making it a versatile solution for various modern communication needs [2].

Limitations:

- The SAPM antenna requires a more complex radiating element (six-cylinder intersection) and an FSS reflector, increasing design and fabrication complexity.
- The FSS reflector and artistic radiating structure require precise manufacturing, potentially increasing complexity and production cost.

- The inclusion of GPR applications with directional far-field patterns may make it less optimal for general UWB communication systems requiring omnidirectional patterns.

The systematic review examines the use of frequency-selective surface (FSS) techniques for gain enhancement in ultra-wideband (UWB) antennas. The focus is on minimizing power loss and interference while achieving directional radiation. The review explores both single-layer and multi-layer FSS reflectors, analysing their impact on antenna performance. Additionally, the paper proposes strategies for FSS miniaturization and methods to maintain a constant gain across UWB frequencies while ensuring that the return loss remains below −10 db. The analysis draws from 41 papers sourced from major research databases, providing a comprehensive look at the advancements and techniques in the field of gain enhancement for UWB antennas using FSS [3].

Limitations:

- Focuses primarily on gain enhancement and directional radiation but does not address other critical performance metrics like group delay.
- FSS techniques often require additional space and complexity (e.g., reflector structures), which may not be suitable for compact or portable applications.
- Provides a theoretical review without presenting experimentally validated results.

1.3 Problem Statements

- Modern wireless communication systems require compact, efficient, and cost-effective antennas capable of operating over an ultra-wideband (UWB) frequency range. However, understanding the principles and fundamentals of coplanar waveguide (CPW)-fed UWB antennas, including their design, operational parameters, and performance characteristics, remains challenging due to the complex interactions between various electromagnetic phenomena.
- Ultra-wideband technology offers significant advantages for high-speed data transmission, low power consumption, and precise localization. Despite these benefits, the integration of UWB in real-world wireless applications faces issues such as interference, signal attenuation, and regulatory constraints, making it necessary to conduct a detailed analysis to identify and overcome these barriers.
- As modern devices become increasingly compact, there is a growing demand for miniaturized antennas without compromising performance. Designing a CPW-fed

UWB antenna that meets size constraints while achieving high efficiency, wide bandwidth, and good impedance matching poses a significant technical challenge.

1.4 Objectives

The primary objective of this project is to explore, analyse, and evaluate the design and performance of a miniaturized coplanar waveguide (CPW)-fed ultra-wideband (UWB) antenna for wireless applications. The report aims to address key challenges in antenna design while emphasizing the importance of miniaturization, wide bandwidth, and efficient performance. The objectives of the report are outlined as follows:

1. To Study the Fundamentals of CPW-Fed UWB Antennas.
2. To Analyse Ultra-Wideband Technology for Wireless Applications.
3. To Design a Miniaturized CPW-Fed UWB Antenna.

Chapter 02

SYSTEM REQUIREMENTS

2.1 Hardware Requirements

FR4 Substrate: FR4 is a widely used material for antenna designs due to its excellent electrical and mechanical properties. FR4's affordability, ease of manufacturing, and adequate performance for low to moderate frequency ranges make it an ideal choice for many practical antenna applications.

- It is a glass-reinforced epoxy laminate material with a dielectric constant of 4.4, which ensures stable signal propagation and adequate insulation.
- The FR4 substrate typically has a thickness of 1.6 mm, which balances mechanical rigidity and electromagnetic performance.
- A feeder line with a characteristic impedance of 50 Ω is commonly used to ensure proper matching between the antenna and the connected transmission line or device, minimizing signal reflection and power loss.

2.2 Software Requirements

CST Studio Suite (Computer Simulation Technology) : is a 3D electromagnetic simulation software, offering tools to analyse and optimize designs across a wide frequency range. It is widely used in industries such as telecommunications, aerospace, automotive, and healthcare for applications like antenna design, electromagnetic compatibility (EMC) analysis, and signal integrity optimization.

CST combines high accuracy with user-friendly workflows, enabling engineers to perform simulations efficiently and reduce the need for physical prototypes, thus saving time and costs. Its modules support diverse physics domains, including low-frequency, high-frequency, and thermal analysis, making it a versatile solution for electromagnetic challenges. The software specification is as follows;

- Processor: Intel Xeon (recommended).
- Graphics: OpenGL-compatible hardware.
- Memory: Minimum 4 GB RAM.
- Storage: 30 GB for CST Suite, 2 GB for Antenna Studio Suite.
- Platform: 64-bit PC hardware.

Chapter 03

SYSTEM DESIGN AND IMPLEMENTATION

3.1 Antenna Design

The different design steps of the proposed antenna are shown in Fig. 3.1. Initially, the basic design consists of a simple rectangular monopole radiator excited by a CPW feedline (ANT I). In the next step, the rectangular radiator is truncated from its upper and lower sides in order to keep the return loss close to −10dB (ANT II). In step three, a semi-circular shape is added in the ANT II in order to keep some portion of the UWB band below −10dB (ANT III). In step four, a C-shaped resonator is introduced to make the design jug shaped which keeps the whole UWB band below −10dB perfectly (ANT IV).

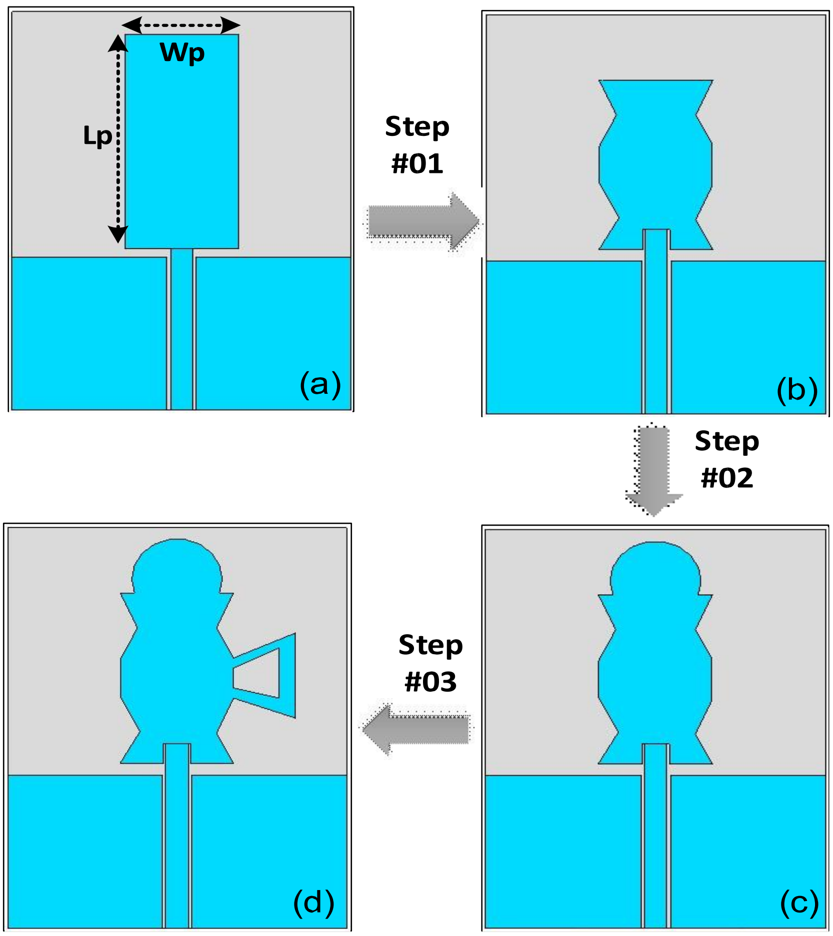


Figure 3.1: Design Steps of the proposed ultra-wideband Antenna

(a) rectangular patch only (ANT I) (b) Truncated patch (ANT II)

(c) Addition of semi-circular patch (ANT III) (d) Proposed design (ANT IV)

A schematic illustration of the orthogonal antenna is shown in Fig. 3.2. The suggested UWB antenna's design consists of a jug-shaped patch with a handle on the right side of the radiator. Because of the antenna's Jug-like shaped construction, it has a larger bandwidth. A coplanar waveguide (CPW) microstrip line feeds the patch.

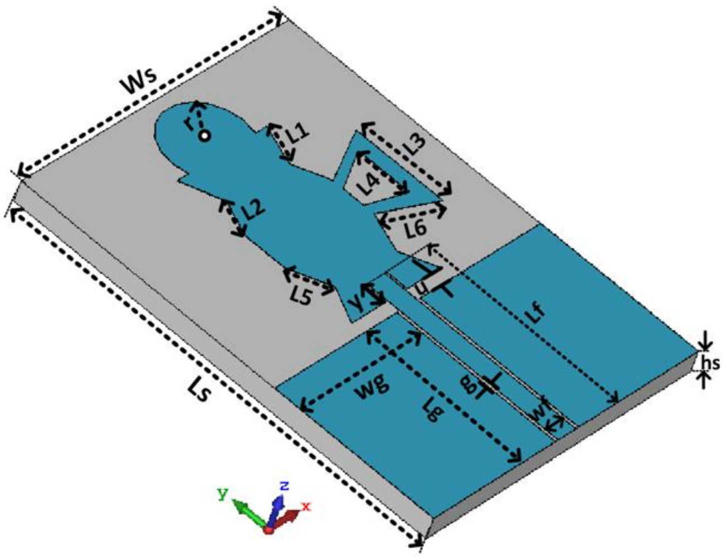


Figure 3.2: Schematic diagram of the proposed ultra-wideband Antenna

The antenna elements are made on an inexpensive FR-4 substrate with a loss tangent (tan δ) of 0.025 and a relative permittivity (ε_r) of 4.4. This antenna design is simulated in the computer simulation technology (CST-2019) Software. The antenna's optimized design parameters are listed in Table 3.1.

Table 3.1: Different design parameters of the proposed UWB antenna

Parameters	Values (mm)	Parameters	Values (mm)
Ls	31.68	Ws	23.76
Lf	14.65	Wf	1.58
Lg	12.14	Wg	10.85
L1	3.02	L2	2.75
L3	6.73	L4	3.96
L5	3.05	L6	4.64
g	0.24	Y	1.58
u	0.95	hs	1.6

3.2 Flowchart

The following is a description of the patch antenna's design process: The primary antenna design includes a 50Ω-CPW feedline, a jug-shaped patch, and a ground plane. (ANT I) depicted in Fig. 1(a). The length and width of the patch are computed using equations (1) and (2) below.

$$Wp = \frac{\lambda_o}{2(\sqrt{0.5(\varepsilon_r + 1)}} \quad \text{--------- (01)}$$

In (1), ε_r and λ_o are the substrate's relative permittivity and free-space wavelength at the operating frequency. The optimal choice of *Wp* results in an excellent impedance matching. The patch's length can be calculated using;

$$Lp = \frac{c_o}{2f_o\sqrt{\varepsilon_{eff}}} - 2\Delta L_p \quad \text{--------- (02)}$$

where c_o, $1L_p$, and ε_{eff} are the speed of light, change in the size of the patch owing to its fringing effect, and the effective dielectric constant, respectively. The effective relative permittivity can be calculated as follows;

$$\varepsilon_{eff} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left(\frac{1}{\sqrt{1 + 12\frac{hs}{W_p}}}\right) \quad \text{--------- (03)}$$

where 'h_{sub}' is the height of the substrate. Finally, the change in the size of the patch, due to its fringing effects, can be calculated using following equation;

$$\Delta Lp = 0.421\mathrm{h}_s \frac{(\varepsilon_{eff} + 0.300)(\frac{W_p}{hs} + 0.264)}{(\varepsilon_{eff} - 0.258)(\frac{W_p}{hs} + 0.813)} \quad \text{--------- (04)}$$

The rectangular patch's starting parameters are *Lp* = 16 mm and *Wp* = 10 mm, based on ε_r = 4.3 and hs = 1.6mm in Equations (1)-(4).

The flowchart illustrates a systematic approach for designing and optimizing a UWB-CPW (Ultra-Wideband Coplanar Waveguide) antenna. Fig 3.3 shows a comparison of the S_{11} for the design steps. The process starts with setting technical specifications like bandwidth and substrate material, followed by iterative design, simulation, and optimization steps.

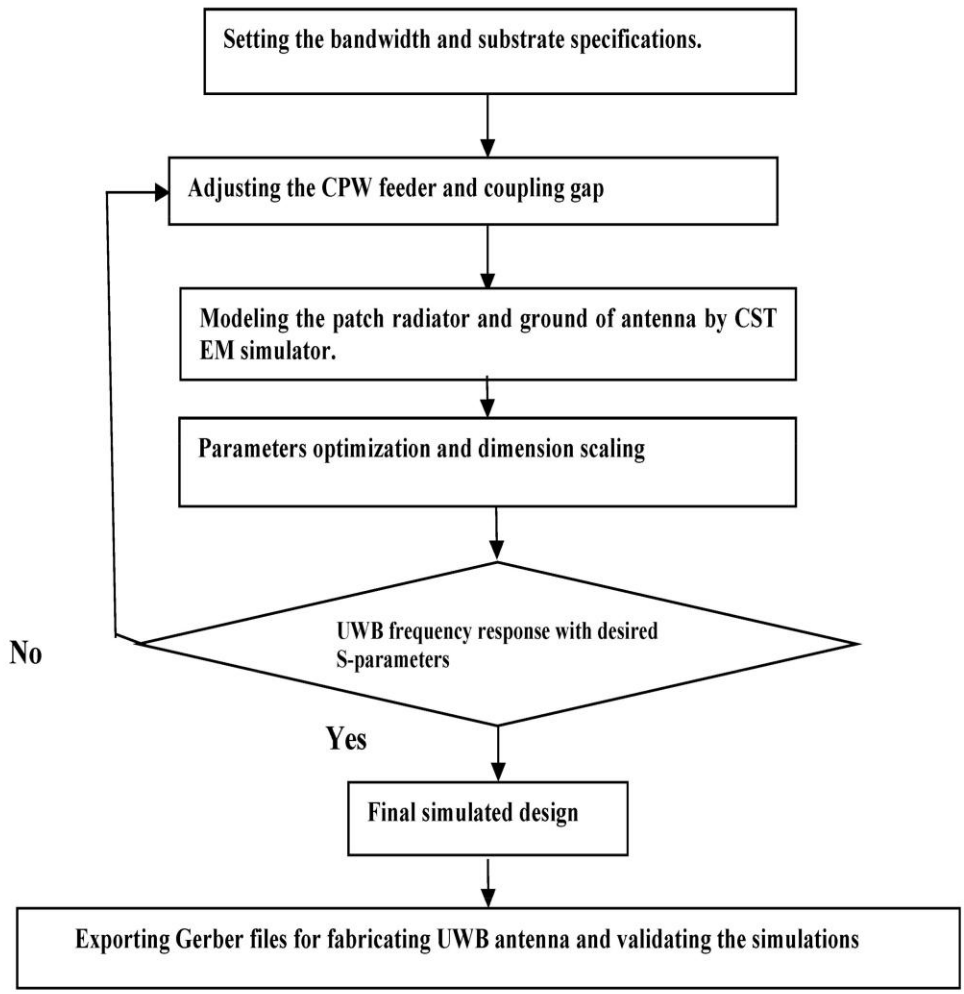


Figure 3.3: Procedural steps for design and optimizing of S_{11} responses of the UWB-CPW antenna

Key parameters like S11 (reflection coefficient) are checked to ensure the antenna meets UWB standards. Once the design satisfies performance criteria, it is prepared for physical fabrication. This method ensures a structured workflow, leveraging simulation tools to reduce trial-and-error efforts during the physical prototyping stage.

Chapter 04

RESULTS AND DISCUSSIONS

4.1 Simulation Result

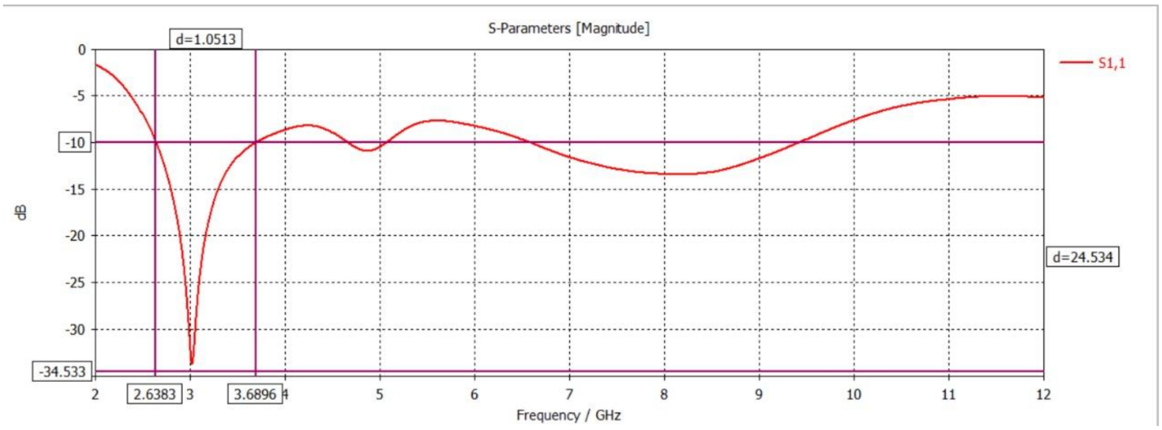
The simulation of the S_{11} parameter in CST Studio Suite involves the evaluation of the antenna's reflection coefficient across a range of frequencies to analyse its impedance matching and bandwidth performance. CST Studio Suite uses computational electromagnetic (EM) methods like the Finite Integration Technique (FIT) or Frequency Domain Solver to solve Maxwell's equations and simulate the antenna's behaviour in its designed environment.

To simulate the S_{11} parameter, the antenna model is first designed with precise geometry, including the CPW feed structure and substrate properties. The material parameters, such as the dielectric constant and loss tangent of the substrate, are assigned to ensure accurate modelling. Ports are then defined at the antenna's feed point, typically as waveguide ports, to simulate the input signal. The simulation setup involves specifying the frequency range (e.g., 2 GHz to 12 GHz for UWB applications) and defining mesh properties for high-resolution results.

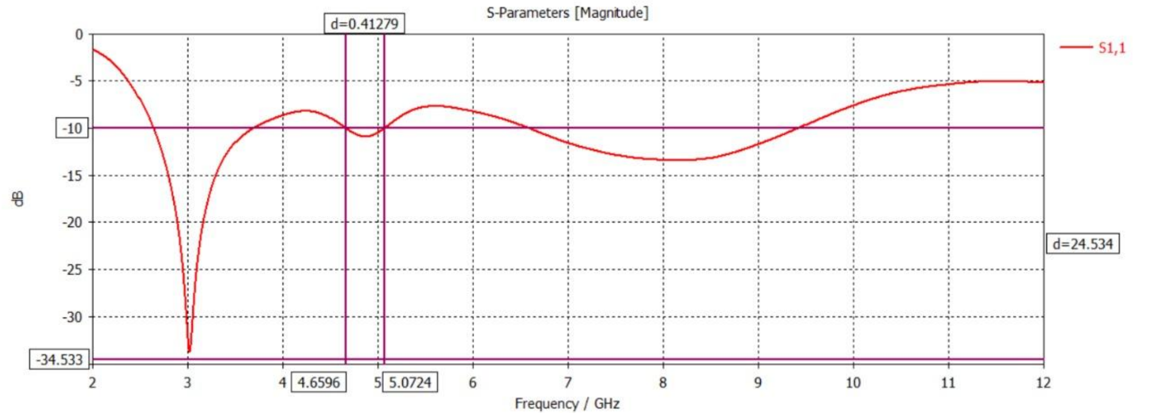
During the simulation, CST calculates the scattering parameters by solving the EM fields at the defined port and the antenna structure. The S_{11} parameter represents the fraction of input power reflected back at the port, indicating how well the antenna is matched to the feed line impedance (typically 50 ohms). After the simulation, CST generates an S_{11} vs. frequency plot, as shown in the figure, where resonance points (deep dips) indicate frequencies with minimal reflection and optimal power transfer. This data helps evaluate the antenna's bandwidth, resonance frequencies, and overall performance, ensuring it meets design requirements for applications like UWB communication systems.

4.1.1 Return loss

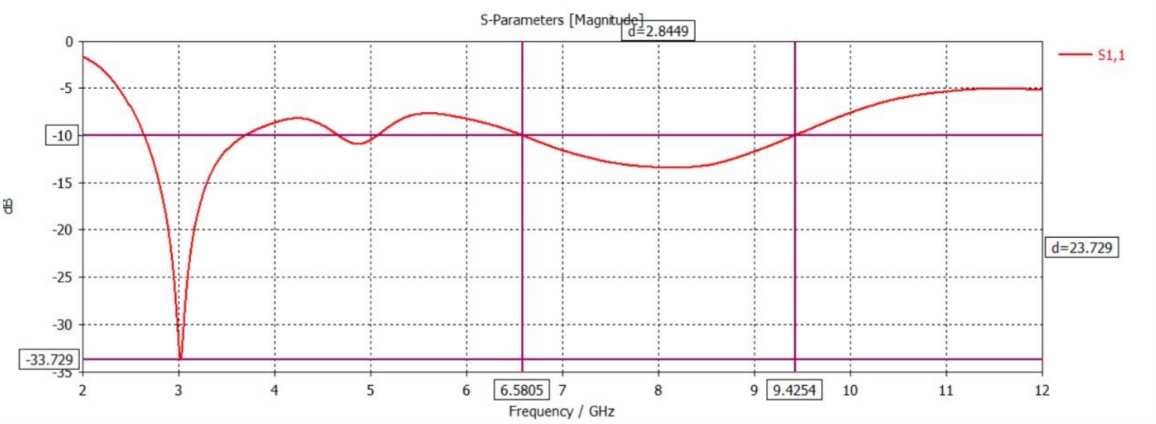
Return loss in antennas is a measure of the efficiency with which the antenna radiates the power delivered to it. It quantifies the mismatch between the antenna and the transmission line, expressed in decibels (dB). A higher return loss indicates a better match, meaning less power is reflected back to the source and more is transmitted or received by the antenna. Ideally, an antenna should have a return loss of -10 dB or better, which corresponds to a Voltage Standing Wave Ratio (VSWR) of 2:1 or lower.



(a)



(b)



(c)

Figure 4.1 (a) Return loss of the frequency band of 2.6383-3.6896GHz of the proposed antenna **(b)** Return loss of the frequency band of 4.6596-5.0724GHz of the proposed antenna, **(c)** Return loss of the frequency band of 6.5805-9.4254GHz of the proposed antenna

The antenna operates over three distinct frequency bands: 2.6383–3.6896 GHz as shown in figure 4.1 (a), 4.6596–5.0724 GHz as shown in figure 4.1 (b), and 6.5805–9.4254 GHz, as shown in figure 4.1 (c) making it versatile for multi-band and ultra-wideband (UWB) applications. It can handle frequencies ranging from 2.6383 GHz to

9.4254 GHz, enabling diverse communication, radar, and sensing applications. The compact design, likely using CPW (coplanar waveguide) feed, ensures ease of integration into modern devices.

Frequency Band 1 (2.6383 - 3.6896 GHz):Wi-Fi/WLAN: High-speed data communication,5G Communication: Mid-band services, Radar Systems: Weather and air traffic control, Satellite Communication: C-band services.

Frequency Band 2 (4.6596 - 5.0724 GHz):Wi-Fi (5 GHz Band): High-speed, low-interference communication, Military Radar: Airborne and naval applications, Point-to-Point Communication: High-bandwidth backhaul links, Industrial Applications: ISM band uses.

Frequency Band 3 (6.5805 - 9.4254 GHz):Satellite Communication: X-band military and commercial services, Radar Systems: Weather, maritime, and automotive radars, Earth Observation: Remote sensing and SAR imaging, Wireless Backhaul: High-capacity microwave links.

.4.1.2 Gain

Antenna Gain refers to the measure of how effectively an antenna directs or concentrates radio frequency energy in a specific direction, compared to a hypothetical isotropic antenna that radiates equally in all directions. It is typically expressed in decibels (dB) and provides an indication of the antenna's efficiency in transmitting or receiving signals. Higher gain antennas focus energy more narrowly, improving communication range and signal strength in a particular direction, while lower gain antennas distribute energy more broadly, covering wider areas.

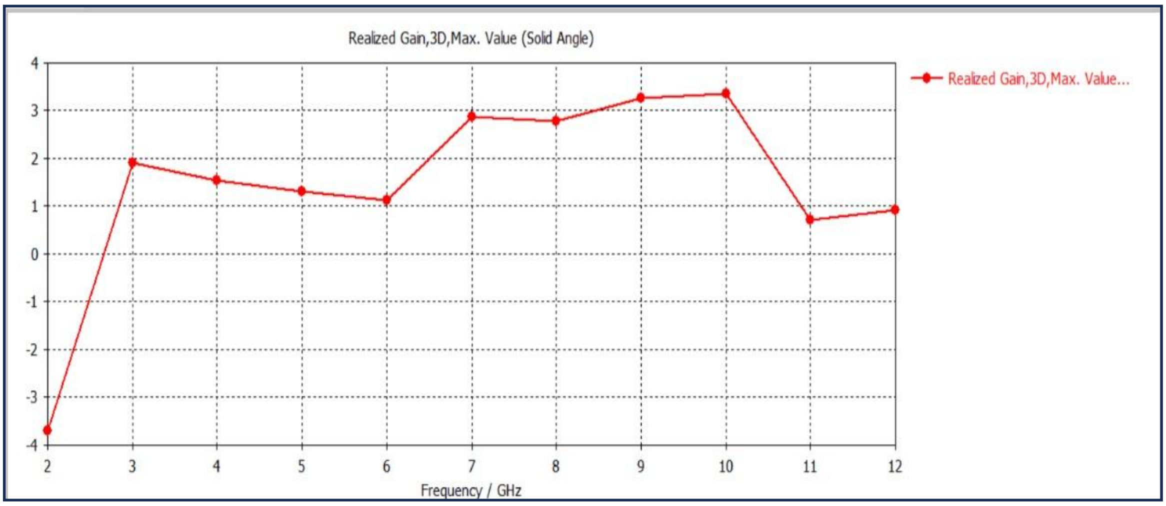


Figure 4.2 : Realised gain

As shown in figure 4.2, graph depicts the realized gain (3D, maximum value) of an antenna across a frequency range from 2 GHz to 12 GHz.

Key Observations:

1. Gain Trend:

- At 2 GHz, the gain is at its lowest, around -4 d.
- The gain rises sharply to about 2 dB at 3 GHz.
- Between 3 GHz and 10 GHz, the gain remains relatively stable, fluctuating slightly but staying above 1 dB, with a peak close to 2.5 db. After 10 GHz, the gain decreases significantly to around 1 dB and then stabilizes.

2. Frequency Range:

- The antenna shows better performance (positive gain) in the frequency range of 3 GHz to 10 GHz.
- Lower gain values are observed at the boundaries of the frequency range (2 GHz and above 10 GHz).

4.1.3 Efficiency

The efficiency of an antenna is a measure of how effectively it converts input power into radiated electromagnetic waves. It is typically expressed as a percentage and is influenced by factors such as resistive losses, impedance mismatch, and material properties. High-efficiency antennas radiate most of the input power, minimizing energy losses in the form of heat or reflected signals. Efficiency is critical in applications where power conservation is important, such as in portable devices, satellite communication, and low-power wireless systems. To achieve optimal efficiency, careful design considerations like proper impedance matching, low-loss materials, and minimizing conductor

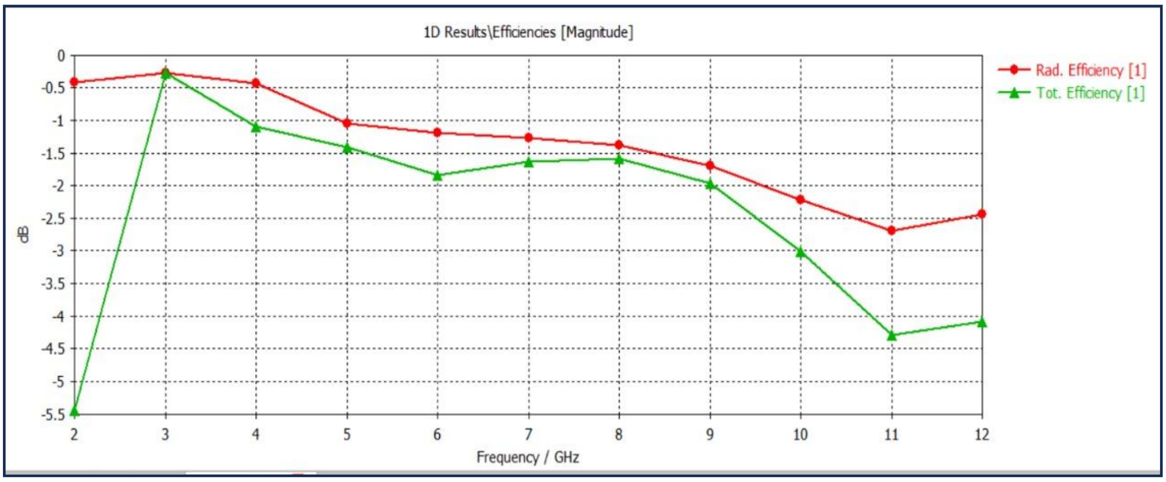


Figure 4.3 : Efficiency in antenna

Figure 4.3 represents the radiation efficiency (red line) and total efficiency (green line) of an antenna over a frequency range of 2 GHz to 12 GHz, measured in decibels (dB).

Key Observations:

1. Radiation Efficiency:

- The radiation efficiency remains relatively stable across the frequency range, fluctuating around -0.5 dB to -1.5 db.
- This indicates minimal losses due to material or design issues in converting input power to radiated energy.

2. Total Efficiency:

- Total efficiency starts at -5.5 dB at 2 GHz, improves sharply to around -1 dB near 3 GHz, and gradually decreases beyond 6 GHz, reaching approximately -4 dB at 12 GHz.
- This decrease suggests increasing losses due to impedance mismatch, material absorption, or other factors at higher frequencies.

3. Performance Across Frequencies:

- Between 3 GHz and 6 GHz, the total efficiency is relatively high, making this the optimal frequency range for the antenna's operation.
- The divergence between radiation and total efficiency increases at higher frequencies, indicating that factors other than radiation losses (e.g., reflection or conduction losses) contribute to reduced total efficiency.

4.1.4 Field Energy

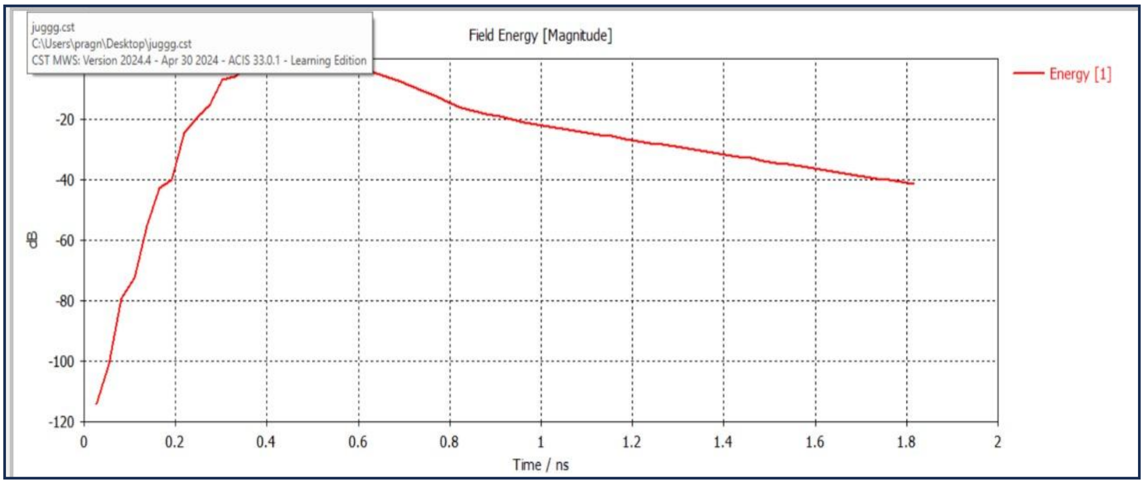


Figure 4.4: Field energy of an antenna

The field energy of an antenna refers to the energy stored and radiated by the electromagnetic fields generated during its operation. This energy can be classified into two components: stored energy and radiated energy. Stored energy is associated with reactive fields (electric and magnetic) near the antenna, which do not contribute to radiation but are necessary for sustaining oscillations. Radiated energy, on the other hand, is the energy carried away by electromagnetic waves into free space.

Figure 4.4 shows the field energy of an antenna as a function of time, measured in nanoseconds (ns). Here's the analysis:

Key Observations:

1. Initial Rise:

- The field energy starts at a very low level (approximately -120 dB) at 0 ns and rapidly increases, reaching its peak around 0.5 ns.
- This indicates the initial buildup of electromagnetic energy as the antenna begins to transmit or receive signals.

2. Peak Field Energy:

- The maximum field energy is observed between 0.4 and 0.6 ns, suggesting efficient energy transfer and storage during this interval.
- This peak represents the antenna's maximum capacity to store and radiate electromagnetic energy.

3. Energy Decay:

- After reaching the peak, the field energy gradually decreases over time, eventually stabilizing at a lower level beyond 1.8 ns.
- This decay reflects the dissipation of stored energy, either radiated into free space or absorbed by the antenna system.

4. Performance Indicators:

- The rapid rise and gradual decay of field energy indicate that the antenna efficiently handles energy transfer.
- The stability of the decay curve beyond 1 ns suggests controlled energy dissipation, essential for consistent antenna performance.

4.1.5 Voltage Standing Wave Ratio (VSWR)

The Voltage Standing Wave Ratio (VSWR) as a function of frequency (GHz) for two different configurations, denoted by the labels "lg=13.5" (red) and "lg=14" (green) as shown in figure 4.5. VSWR is a critical parameter in antenna and transmission line design, representing the efficiency of power transfer between the source and the load. A lower VSWR indicates better impedance matching and more efficient power delivery

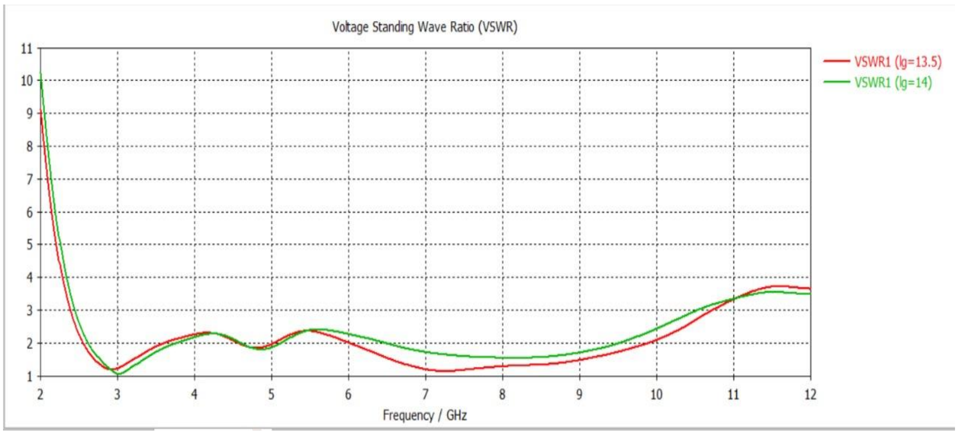


Figure 4.5: VSWR of proposed antenna

Key Observations:

1. Frequency Dependence: The VSWR values vary with frequency, showing resonant points where the VSWR reaches its lowest values (close to 1, indicating near-perfect matching).

2. Multiple Configurations: The two configurations exhibit similar trends but differ slightly in their minimum VSWR values and the frequencies at which these minima occur.

3. Performance Comparison: For frequencies below 3 GHz, both configurations show higher VSWR values, indicating poorer matching, while better matching is achieved in the mid-frequency range (around 3-6 GHz).

4. Impact of Parameter Variation: The parameter "lg" (possibly related to physical or electrical length) has a noticeable influence on the VSWR response, particularly around the resonant frequencies.

4.1.6 Z – Parameter

Z - parameters describe the relationship between voltages and currents in multi-port network systems in terms of impedance as shown in figure 4.6. For a single port, $Z_{1,1}$ represents the input impedance. The figure displays the impedance magnitude ($Z_{1,1}$) versus frequency for two different cases (denoted as lg = 13.5 and lg = 14) in GHz. This is commonly observed in high-frequency circuit analysis, such as transmission lines, antennas, or microwave components.

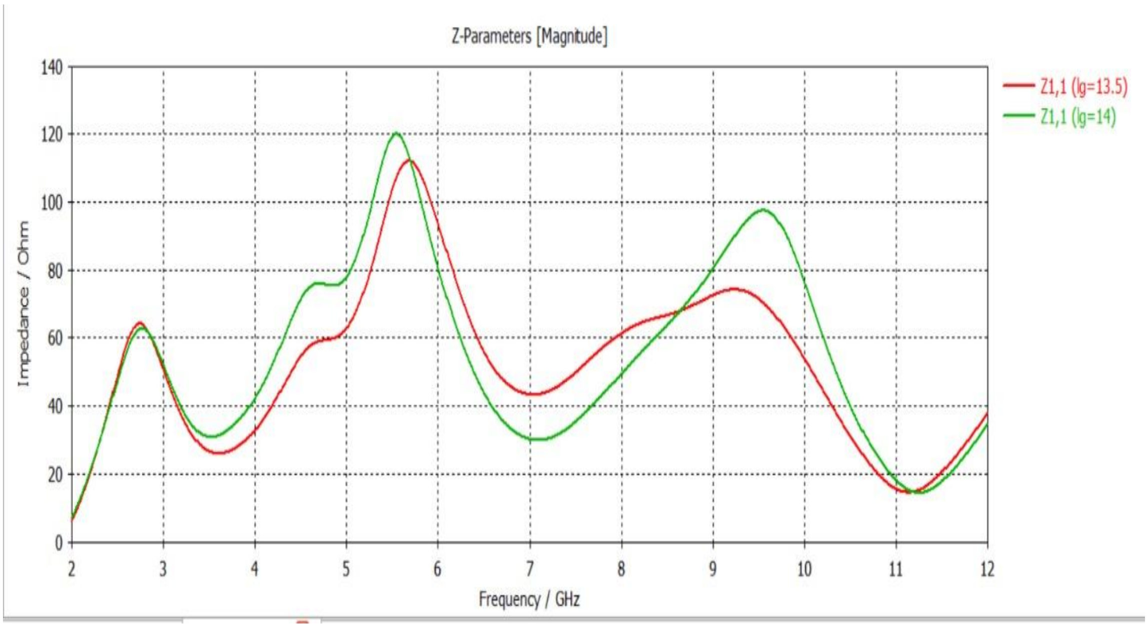


Figure 4.6 : Z parameter of proposed antenna

Key Observations:

1. Resonance Peaks : Multiple resonance peaks are visible in both curves, corresponding to frequencies where the impedance magnitude is highest. These peaks indicate frequencies at which the system naturally oscillates.

2. Effect of lg : Comparing lg = 13.5 (red curve) and lg = 14 (green curve);

- The resonance and anti-resonance frequencies shift slightly due to the change in lg.
- The green curve (lg = 14) exhibits slightly higher peaks at certain frequencies, suggesting increased impedance at those resonances.

4.1.7 Far Field in 3D

The figures 4.7(a), 4.7(b), 4.7(c) depict the far-field radiation patterns of an antenna at frequencies of 3 GHz, 8.05 GHz, and 4.95 GHz, simulated using CST Studio Suite. At 3 GHz, the antenna demonstrates a directivity of 2.161 dB, with a total efficiency of -0.2676 dB, and a gain distribution ranging from -37.8 dB to 2.16 dB, showing a symmetrical pattern concentrated along the central axis. At 8.05 GHz, the antenna exhibits improved directivity of 4.334 dB, though with reduced total efficiency of -1.598 dB, indicating more losses. The gain distribution spans from -35.7 dB to 4.33 dB, with energy focused more directionally. At 4.95 GHz, the antenna achieves a moderate directivity of 2.768 dB and a total efficiency of -1.355 dB, with gain ranging from -37.2 dB to 2.77 dB, maintaining a relatively symmetrical pattern with concentrated intensity at the centre. These results illustrate the variation in antenna performance across different frequencies.

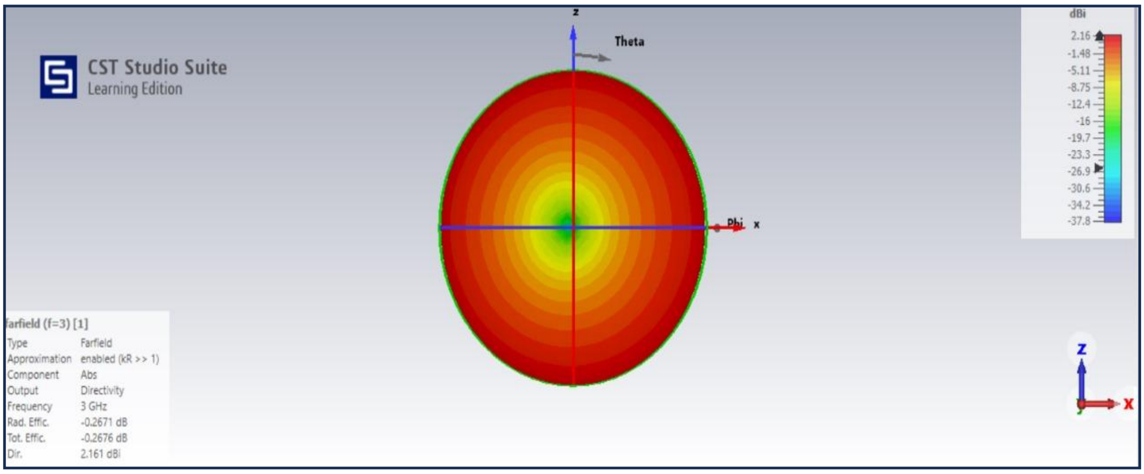


Figure 4.7 (a) : Far Field at 3GHz

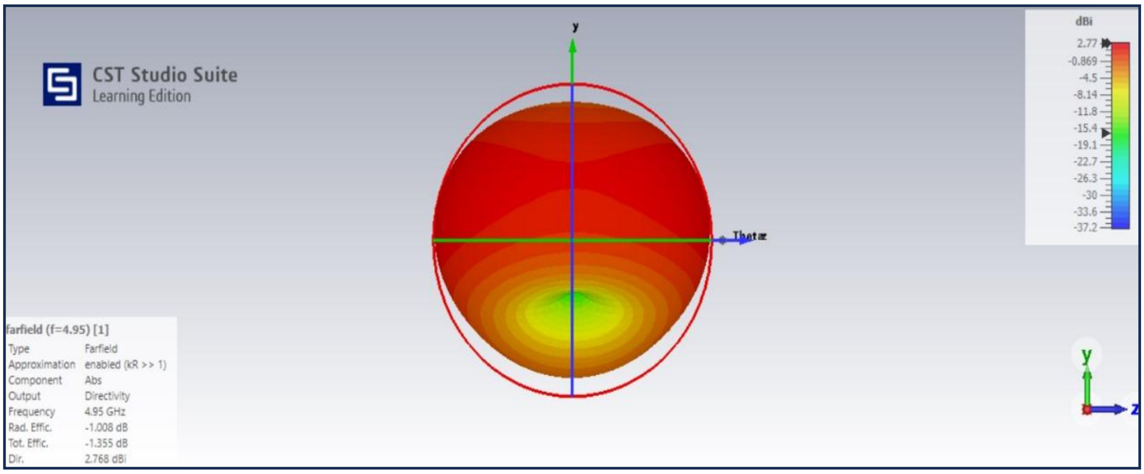


Figure 4.7 (b) : Far Field at 4.95GHz

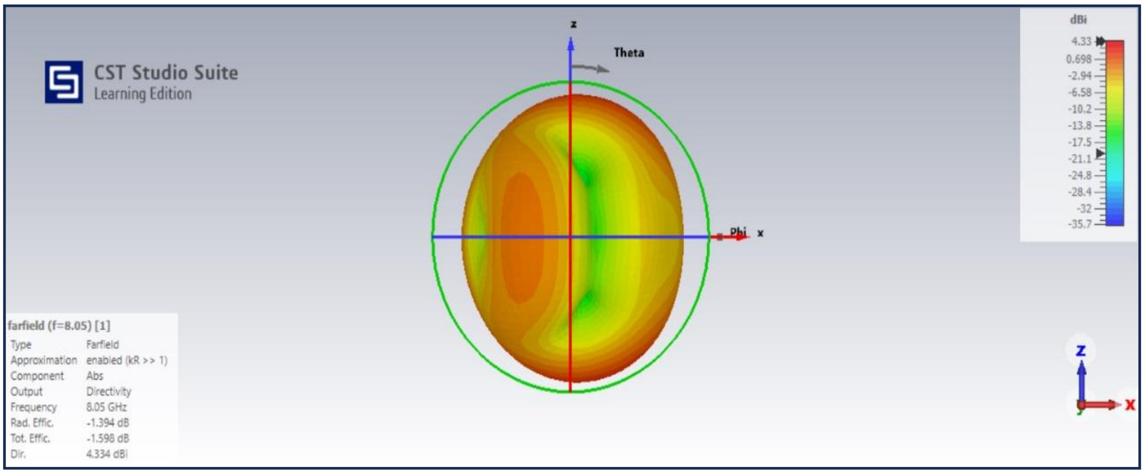


Figure 4.7 (c) : Far Field at 8.05 GHz

4.1.8 Far Field in Polar Form

The far-field directivity plots clearly demonstrate how frequency variation influences antenna performance, particularly in terms of directivity, beam width, and side lobe suppression. At 8.05 GHz, the antenna exhibits a main lobe magnitude of 4.28 dB, with the main lobe directed at 141.0°. The angular width at 3 dB is relatively broad at 171.0°, indicating a wider beam pattern. Side lobe levels are suppressed to -8.9 dB, which suggests moderate control over unwanted radiations. At 4.95 GHz, the main lobe magnitude decreases to 2.69 dB, while the direction shifts to 167.0°, and the angular width narrows to 93.2°. This narrowing reflects an improvement in directivity as the beam becomes more concentrated. At 3 GHz, the antenna's main lobe magnitude further reduces to 2.16 dB, with the main lobe aligning directly at 180.0°. The angular width narrows even further to 82.5°, representing a highly focused beam. The suppression of side lobes becomes more pronounced, minimizing interference and emphasizing the main radiation direction.

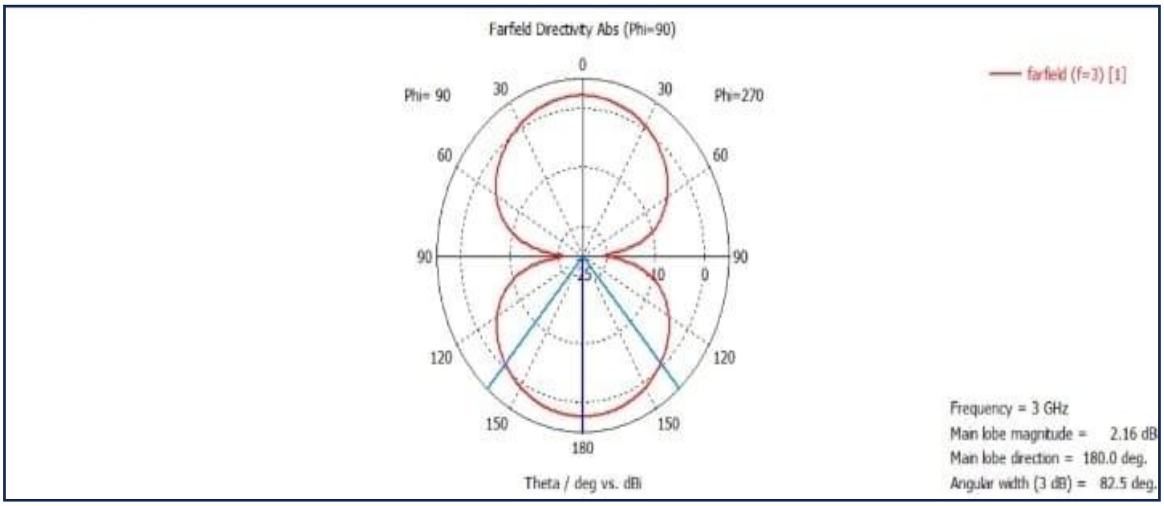


Figure 4.8 (a) : Far Field in Polar Form at 3 GHz

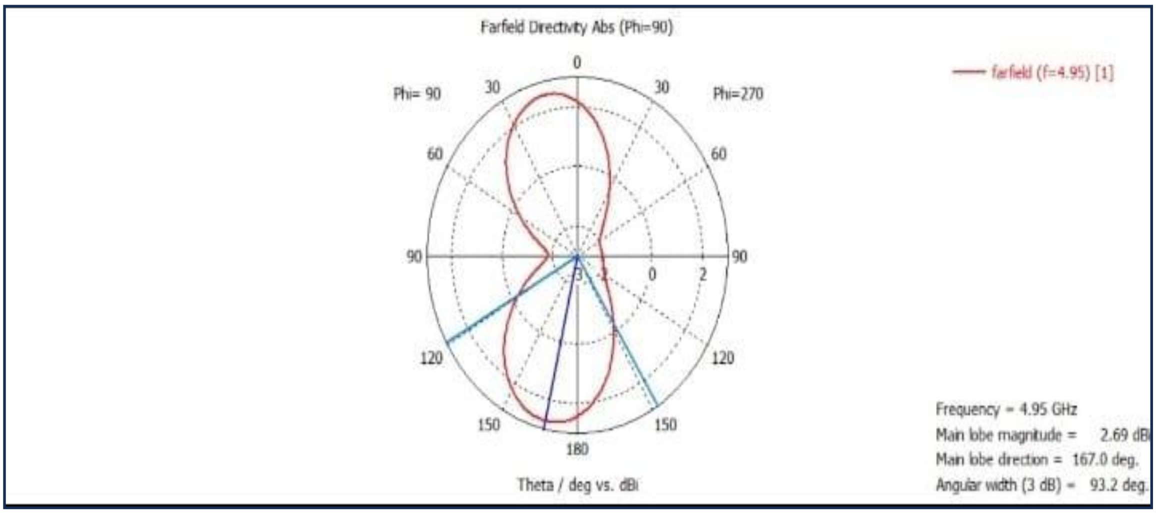


Figure 4.8 (b) : Far Field in Polar Form at 4.95 GHz

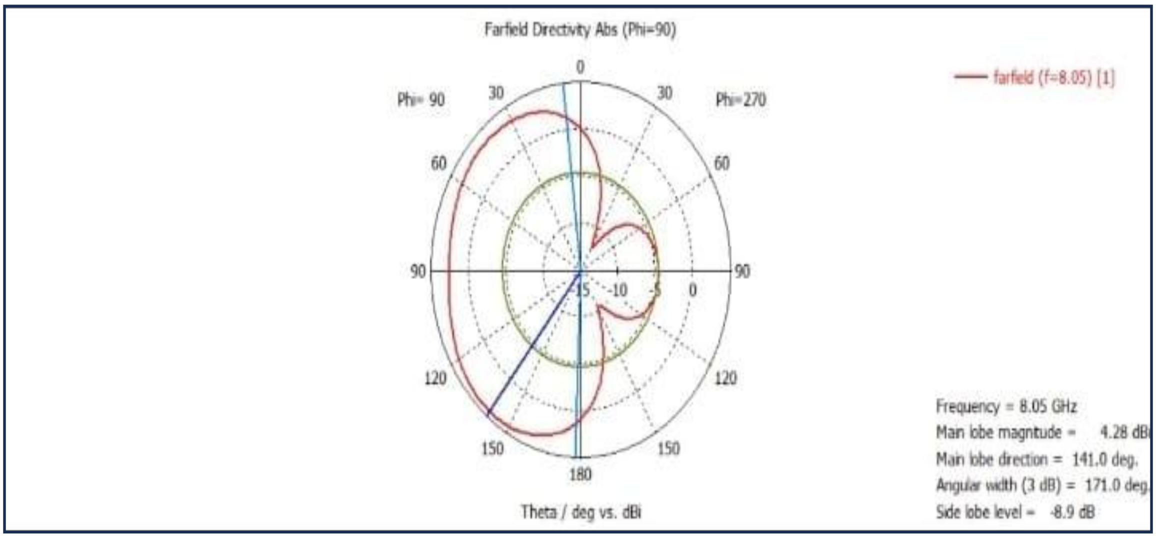


Figure 4.8 (c) : Far Field in Polar format 8.05GHz

4.1.9 E-plane of proposed Antenna

The CPW-fed UWB antenna demonstrates efficient operation across the analysed frequencies (3 GHz, 4.95 GHz, and 8.05 GHz), with the electric field intensity increasing progressively from 2306.5 V/m at 3 GHz to 2814.7 V/m at 8.05 GHz. At 3 GHz, the field distribution is symmetrical and concentrated near the feedline, indicating good resonance at the lower end of the UWB range. At 4.95 GHz, the field spreads more uniformly, suggesting optimal radiation efficiency and impedance matching at this mid-band frequency. At 8.05 GHz, the field intensifies near the radiating edges, potentially enhancing directivity at the higher end. Overall, the antenna exhibits consistent performance, making it well-suited for UWB applications.

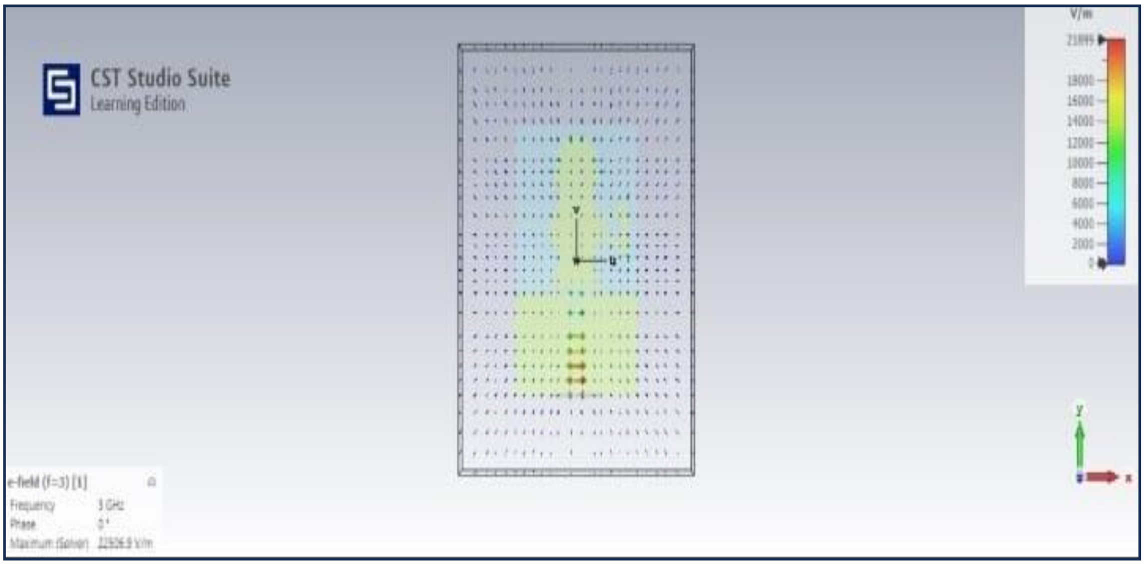


Figure 4.9 (a): E-plane at 3 GHz

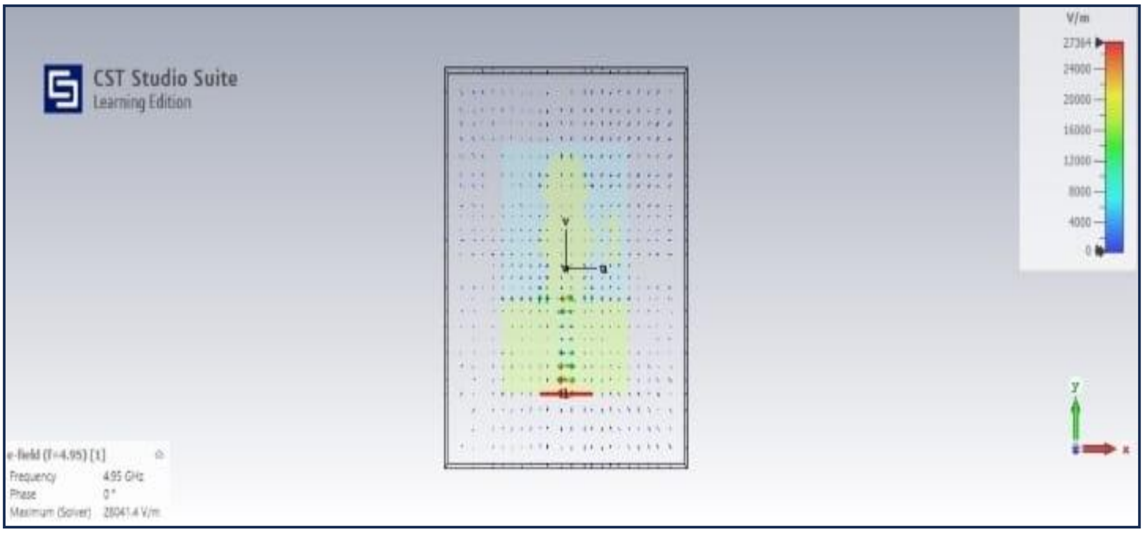


Figure 4.9 (b): E-plane at 4.95 GHz

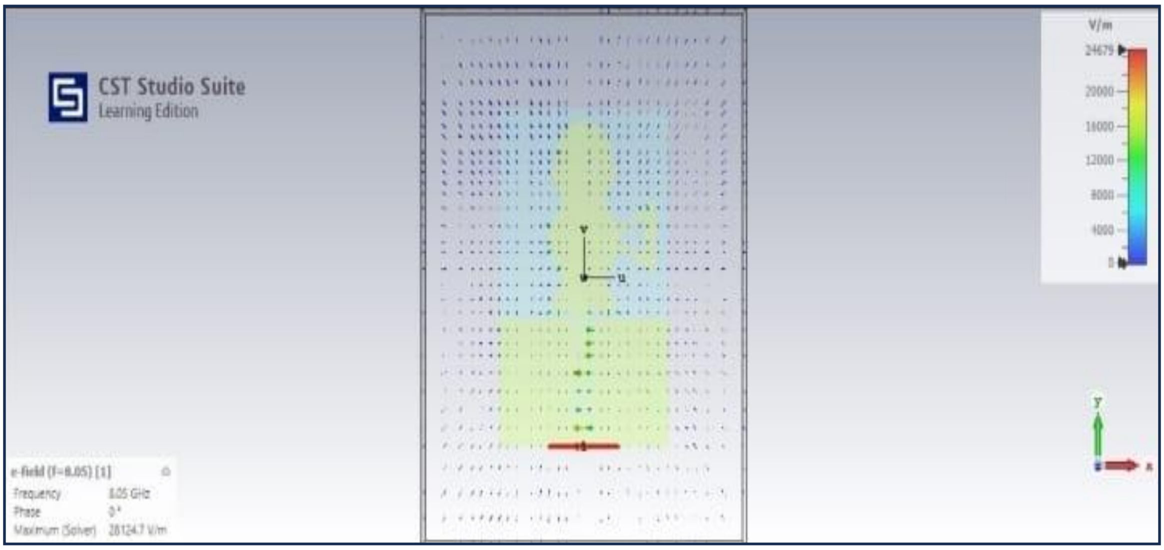


Figure 4.9 (c): E-plane at 8.05 GHz

4.1.10 H-plane of proposed antenna

The figure 4.10(a),(b),(c) illustrate the magnetic field (H-field) distributions at different frequencies. At 4.95 GHz, the maximum H-field intensity is 116.163 A/m, with the field vectors and colour map showing the intensity variation, concentrated in specific regions. At 8.05 GHz, the maximum H-field intensity increases to 125.784 A/m, with the higher frequency resulting in more dynamic field variations and tighter vector distributions. At 3 GHz, the maximum H-field intensity is 88.7245 A/m, with a broader and less concentrated field distribution due to the lower frequency. In all cases, the phase is set to 0°, indicating static field visualization.

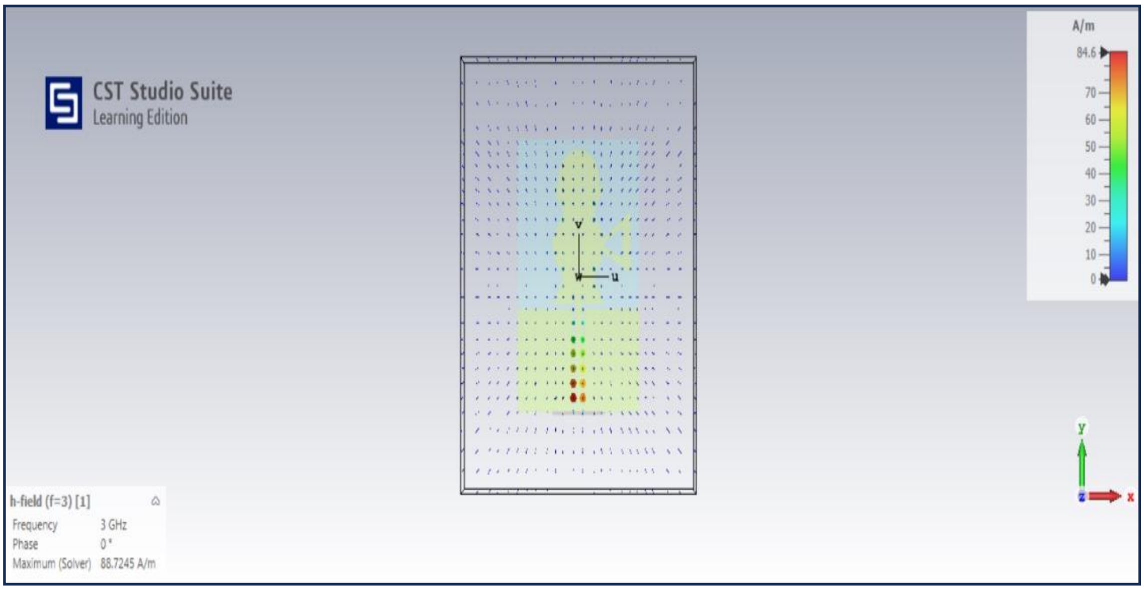


Figure 4.10(a): H-plane at 3GHz

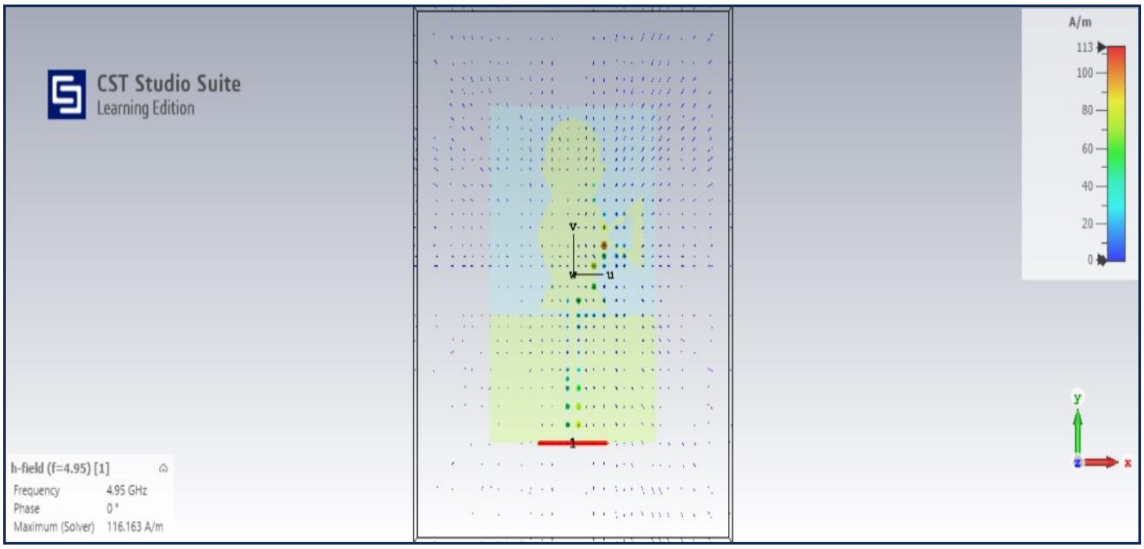


Figure 4.10(b): H-plane at 4.95GHz

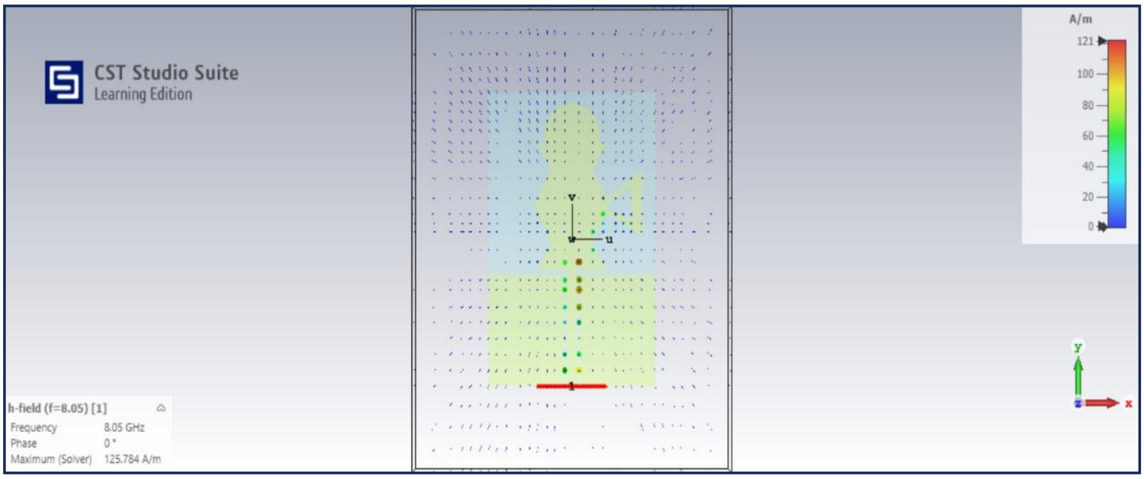


Figure 4.10 (c): H-plane at 8.05GHz

4.1.11 Surface current of proposed antenna

The surface current distribution plots of figure 4.11(a),(b),(c) how the frequency affects the current behaviour on the antenna. At 8.05 GHz, the maximum current density is 121.222 A/m with a phase of 22.5°, showing highly localized current peaks and rapid variations due to the high-frequency behaviour. At 4.95 GHz, the maximum current density decreases slightly to 113.062 A/m, and the phase shifts to 90°. The current distribution becomes less localized, spreading more evenly across the structure, indicating broader resonance. At 3 GHz, the maximum current density further reduces to 84.6316 A/m with a phase of 0°. The distribution exhibits an even broader spread, reflecting the lower frequency's longer wavelength and smoother transitions. These trends highlight the relationship between frequency, current density, and distribution uniformity, with lower frequencies resulting in more distributed and less intense surface currents.

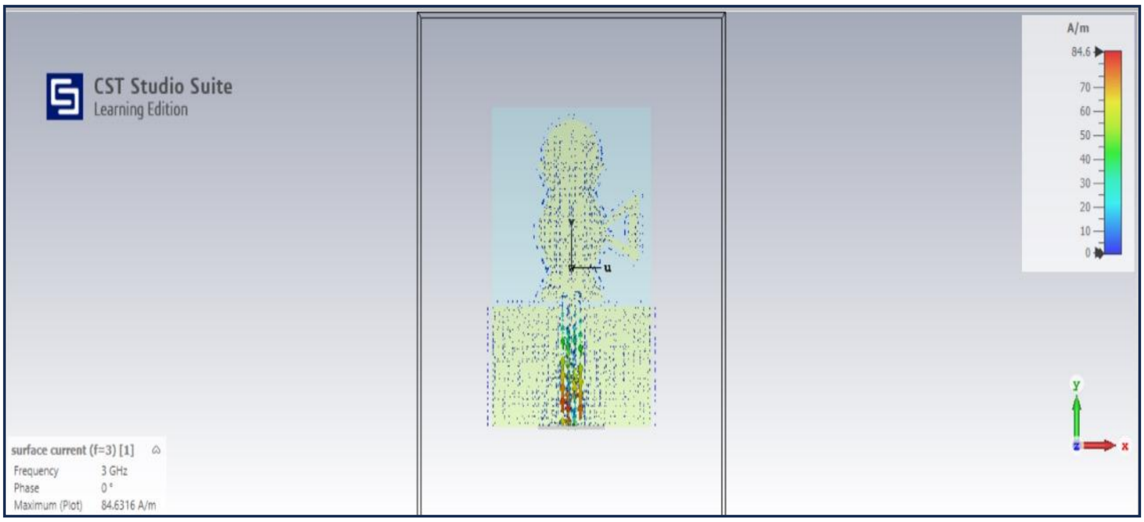


Figure 4.11(a): Surface current at 3GHz

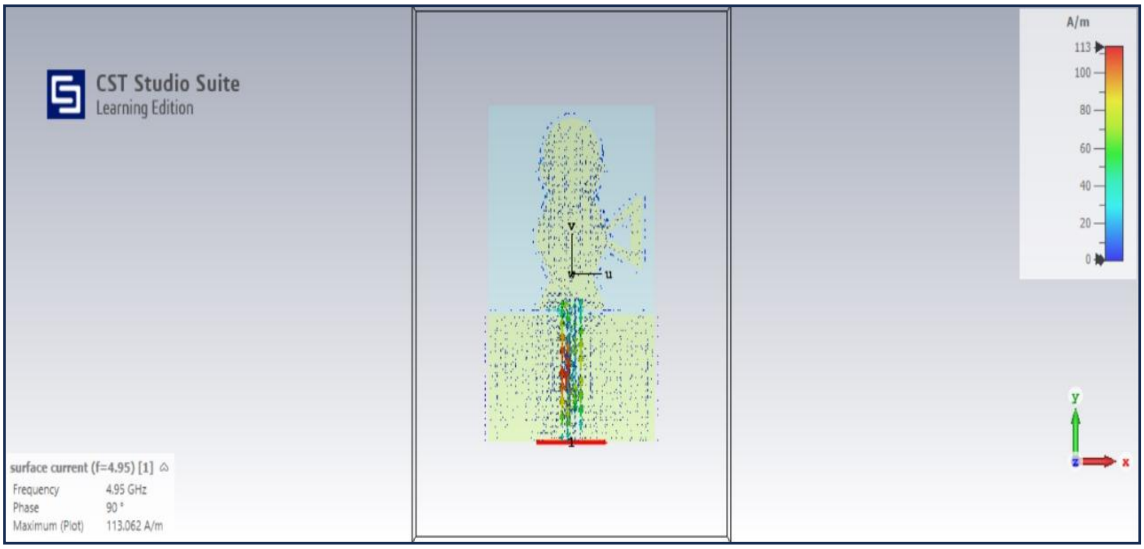


Figure 4.11(b): Surface current at 4.95GHz

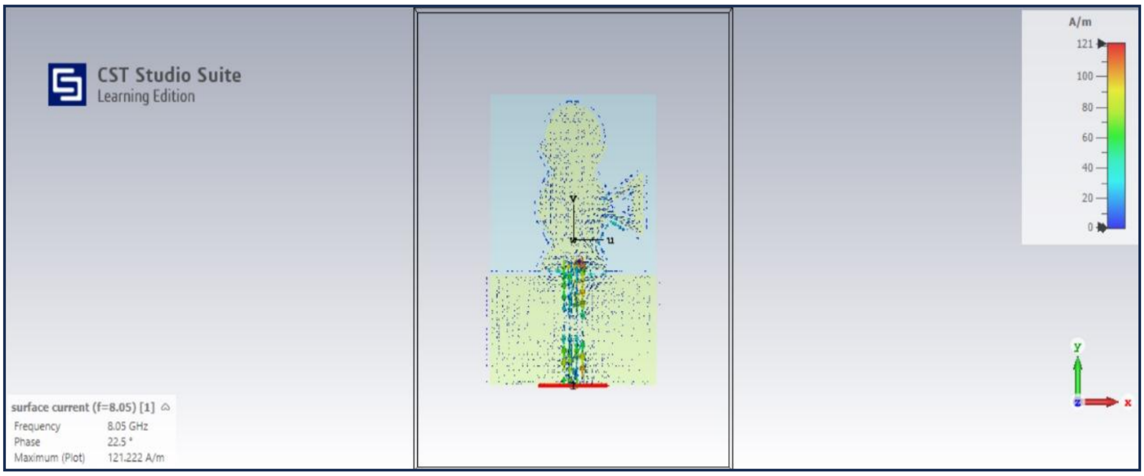


Figure 4.11(c): Surface current at 8.05GHz

4.2 Fabrication

The fabrication process of a miniature CPW-fed UWB antenna begins with the preparation of a suitable substrate, such as FR4 or Rogers RT/Duroid, featuring a copper-clad layer is shown in Figure 4.9. The antenna design is transferred onto the substrate using photolithography or PCB etching. In photolithography, a photoresist layer is applied to the copper surface, followed by aligning a printed mask of the antenna layout.

The substrate is then exposed to UV light, and the exposed areas are etched away using an etchant like ferric chloride, leaving behind the antenna structure. After etching, the PCB is cleaned to remove residual chemicals, and holes are drilled for integrating an SMA connector. Finally, the SMA connector is soldered to the CPW feed for connectivity and testing, completing the fabrication process.

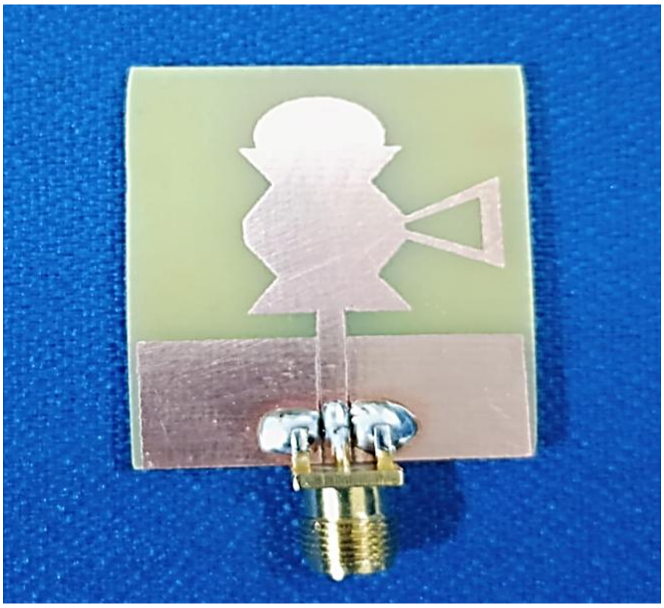


Figure 4.12: Fabricated antenna

4.2.1 Anacova Chamber

The ANACOVA chamber method is a popular approach in the design and optimization of coplanar waveguide (CPW)-fed ultra-wideband (UWB) antennas using FR4 substrate material. This method involves fine-tuning the antenna geometry to achieve desired performance metrics such as impedance matching, radiation efficiency, and wideband characteristics. CPW feeding is particularly advantageous due to its simple planar structure, ease of integration with active components, and reduced losses as shown in Figure 4.10. By employing the ANACOVA technique, designers can systematically analyse and optimize key parameters like the ground plane dimensions, feedline width, and patch shape to enhance the antenna's operational bandwidth and minimize return losses.

Using FR4 as the substrate adds practicality to UWB antenna design, as it is cost-effective, widely available, and exhibits stable electrical properties. However, FR4's relatively high dielectric constant and loss tangent pose challenges in achieving wideband performance. The ANACOVA chamber method addresses these challenges by facilitating an iterative design process that compensates for material limitations, ensuring that the antenna maintains efficient radiation over the entire UWB spectrum (3.1–10.6 GHz). This method, coupled with CPW feeding, results in a compact, lightweight, and efficient UWB antenna, suitable for modern communication and sensing applications.

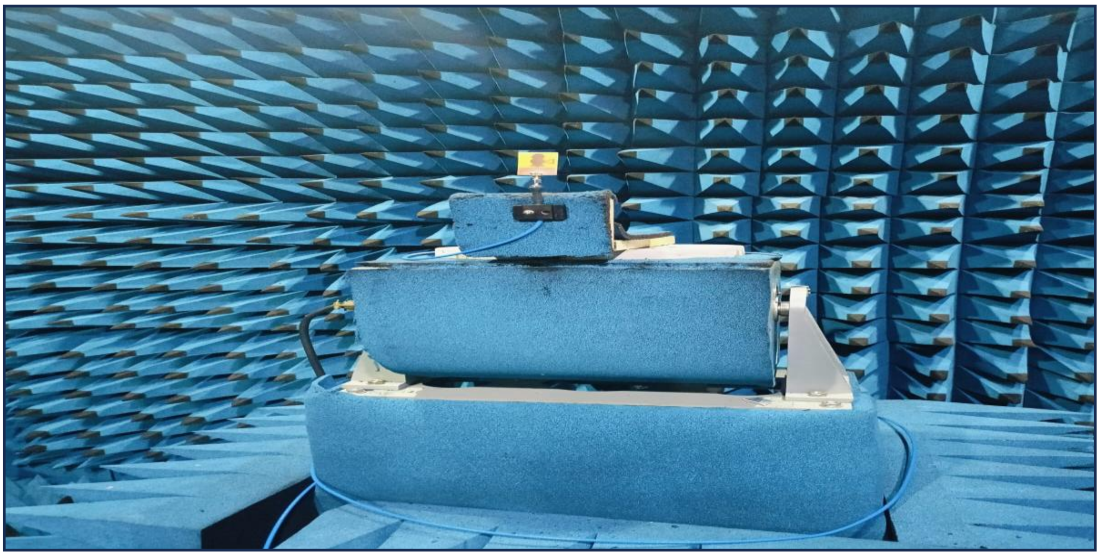


Figure 4.13: ANACOVA Chamber

4.3 Measurement Setup

Antenna measurement involves evaluating key parameters such as resonant frequency, return loss, bandwidth, and impedance matching to ensure optimal performance in communication systems. Using a Vector Network Analyzer (VNA), the reflection coefficient (S_{11}) is measured to determine how effectively the antenna radiates energy at specific frequencies. A good antenna design shows a significant dip in the S_{11} curve at its resonant frequency, indicating minimal power reflection and efficient operation. These measurements are crucial for verifying the antenna's performance against design specifications and ensuring reliable operation in its intended application. The below figure 4.11 illustrates the process of measuring the performance of a fabricated antenna using a Vector Network Analyzer (VNA), specifically the Rohde & Schwarz ZNB 40.

The antenna, connected to the VNA's port via a coaxial cable, is under test to evaluate its key parameters. The displayed graph represents the reflection coefficient (S_{11}), a crucial measure of how much power is reflected back due to impedance mismatch. The dip in the curve (marked by the markers M1 and M2) indicates the resonant frequencies of the antenna, where the return loss is minimized, signifying efficient radiation. The markers also provide numerical values for the resonant frequencies and the associated return loss. This setup allows for detailed analysis of the antenna's resonant frequency, bandwidth (measured as the range of frequencies with return loss below -10 dB), and overall impedance matching, essential for ensuring optimal performance in communication systems. Comparing these measurements with simulated results validates the design and highlights areas for improvement.

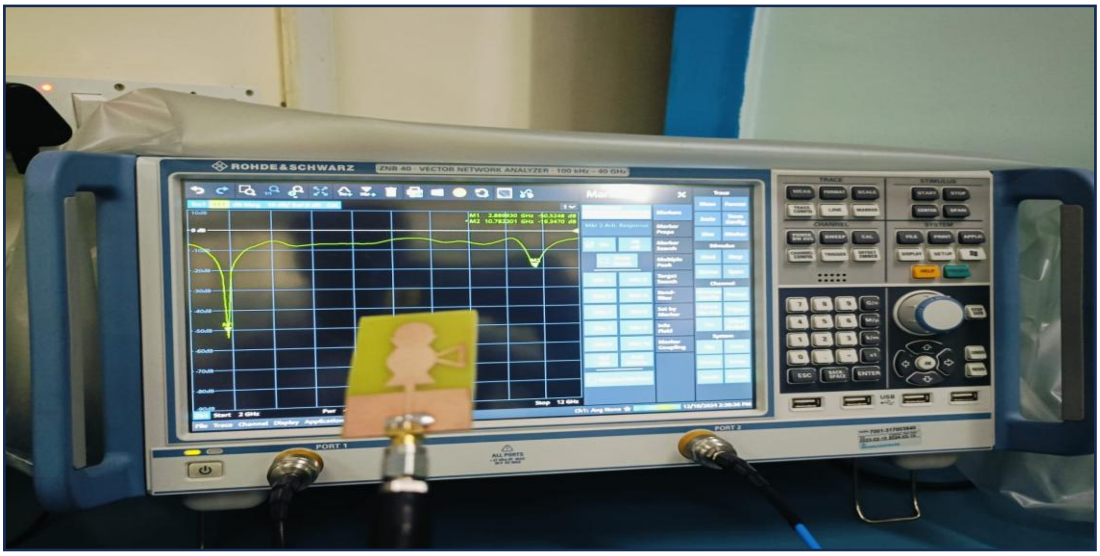


Figure 4. 14: Measurement setup

4.3.1 Reflection coefficient of Fabricated Antenna

The figure 4.15 illustrates an S_{11} magnitude plot (in dB) as a function of frequency, which represents the reflection coefficient of a device or network. The S_{11} parameter measures how much power is reflected back at port 1 relative to the input power.

In this graph, the frequency range spans from 2 GHz to 12 GHz, and the vertical axis shows the reflection magnitude in db. The markers M1 and M2highlight significant points: M1 at 2.88993 GHz shows a reflection magnitude of (-50.5248) dB, indicating minimal reflection and thus a good impedance match at this frequency. Conversely, M2 at 10.782201 GHz has a reflection magnitude of (-19.3470) dB, suggesting higher reflection at this frequency.

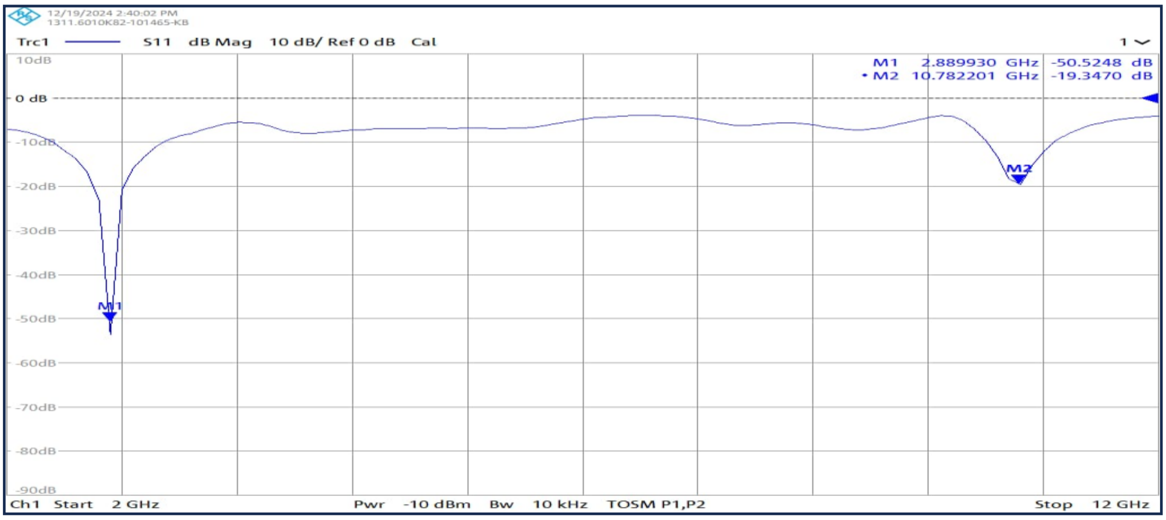


Figure 4.15: Reflection coefficient of fabricated antenna

The dips in the S_{11} plot correspond to frequencies where the device achieves a good impedance match, minimizing signal reflections. Such behaviour is desirable for antennas or transmission lines designed to operate efficiently at specific frequencies. Peaks, or higher values, indicate poor matching, where more power is reflected. This information helps engineers identify resonance frequencies, optimize matching networks, and ensure minimal signal loss. Overall, the S_{11} parameter is a critical metric in RF and microwave engineering, providing insights into the performance and design of components and systems.

4.3.2 Gain of Fabricated Antenna

The figure 4.16 displays an S_{21} magnitude plot (in dB) as a function of frequency, representing the transmission characteristics of a two-port network. In this graph, S_{21}measures how much power is transmitted from port 1 to port 2, with negative values indicating losses. The horizontal axis shows the frequency range (2 GHz to 12 GHz), while the vertical axis represents the transmission magnitude in decibels (dB). Markers M1 and M2 highlight specific frequencies of interest: at M1 (2.88993 GHz), the transmission magnitude is (-17.7443) dB, and at M2 (10.782201 GHz), it is (-30.7406) db. These points may correspond to resonances, attenuations, or cut-off frequencies of the device under test.

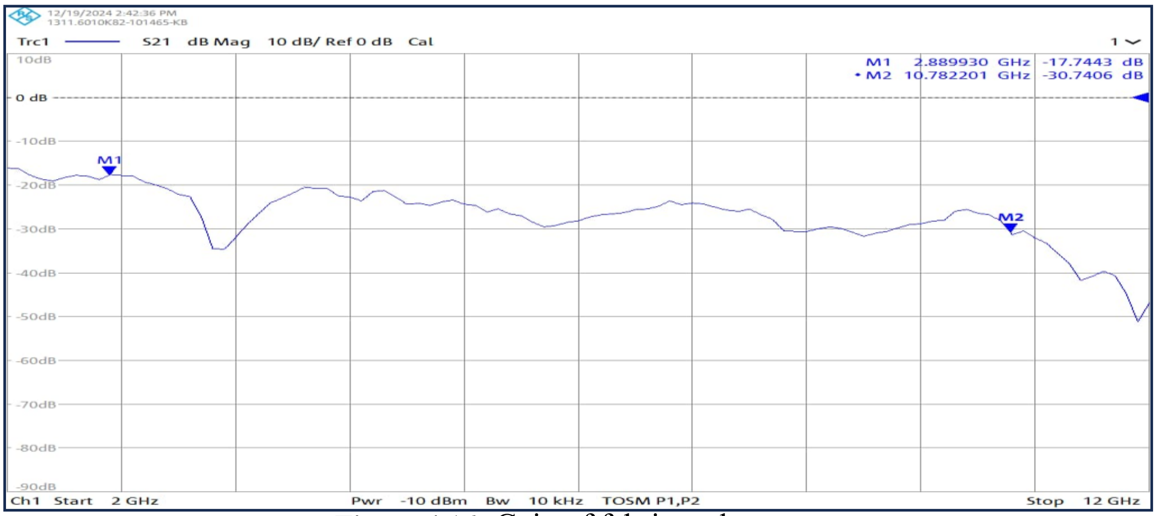


Figure 4.16: Gain of fabricated antenna

The figure 4.16 displays an S_{21} magnitude plot (in dB) as a function of frequency, representing the transmission characteristics of a two-port network. In this graph, S_{21}measures how much power is transmitted from port 1 to port 2, with negative values indicating losses. The horizontal axis shows the frequency range (2 GHz to 12 GHz), while the vertical axis represents the transmission magnitude in decibels (dB). Markers M1 and M2 highlight specific frequencies of interest: at M1 (2.88993 GHz), the transmission magnitude is (-17.7443) dB, and at M2 (10.782201 GHz), it is (-30.7406) db. These points may correspond to resonances, attenuations, or cut-off frequencies of the device under test.

The variations in the curve indicate the frequency-dependent behaviour of the network. Peaks or valleys in the plot signify regions of relatively higher or lower power transmission, respectively. Such plots are crucial for analysing components like filters, amplifiers, or antennas. For example, a steep drop in magnitude at specific frequencies may indicate strong attenuation, aligning with the desired filtering characteristics of a bandpass or notch filter.

4.3.3 Comparison Table

Antennas with higher dielectric constants, like FR4 (εr = 4.4), enable compact designs but have narrower bandwidths, while materials with lower dielectric constants offer wider bandwidths but require larger dimensions. The frequency range varies across designs, with some antennas operating in ultra-wideband (UWB) ranges, suitable for diverse applications such as energy harvesting, RFID, and wearable systems.

Table 4.1 : Comparison between proposed antenna with reported antennas

Ref No.	Size (mm^3)	Dielectric Constant	Frequency Range (GHz)	Max. Gain (dB)	Max. Efficiency (%)	Applications
11	27 x 25 x 1.5	4.4	3.581–13.935	4.7	82	Energy harvesting
14	27 x 33 x 0.787	2.2	3–11 (with 5 notches)	5	40	UWB radio frequency identification (RFID) is used for indoor localization
18	50 x 44.4 x 0.15	4.4	7.2 – 9.2	8	85	Wearable antenna applications
Proposed Antenna	31.68 x 23.76 x 1.6	4.4	Frequency band 1: 2.6383–3.6896	3.34	91.20	S-band radar, satellite communications, industrial Wi-Fi
			Frequency band 2: 4.6596–5.0724			5G mid-band, Wi-Fi 6, C-band satellite uplinks
			Frequency band 3: 6.5805–9.4254			X-band radar, weather monitoring, air traffic control, military surveillance, satellite communications, medical imaging

The table 4.1 provides a comparative analysis of various antenna designs based on parameters such as size, substrate material, dielectric constant, frequency range, gain, efficiency, and applications. The substrate materials, including FR4-epoxy and semi-flexible Rogers RT, play a critical role in determining the dielectric constant, which influences the antenna's bandwidth, impedance matching, and radiation properties.

4.4 Advantages

1. **Broad Impedance Bandwidth:** The antenna covers a wide frequency range (3 GHz to 11 GHz), making it suitable for ultra-wideband applications.
2. **Compact Size:** With a size of 60 × 60 mm², the design is compact and space-efficient, ideal for portable and integrated systems.
3. **Low Group Delay:** The group delay is less than 1.2 ns, ensuring minimal signal distortion and reliable performance.

4. **Efficient Substrate Material The** use of an FR-4 substrate ensures cost-effectiveness and durability while maintaining good electrical performance.
5. **Low Channel Capacity Loss (CCL):** A CCL of less than 0.4 bits/sec/Hz signifies efficient data transmission with minimal loss.
6. **Excellent Reflection Coefficient:** The total active reflection coefficient (TARC) is below −10 dB across the entire bandwidth, indicating strong signal integrity.
7. **Versatility:** The design's performance makes it suitable for various ultra-wideband wireless communication systems and portable devices.

4.5 Disadvantages

1. **Substrate Loss:** The FR-4 substrate has a relatively high loss tangent ($\tan\delta = 0.025$), which may limit efficiency at higher frequencies.
2. **Limited Power Handling:** Thin substrates and compact designs might restrict the antenna's ability to handle high power levels.
3. **Fabrication Challenges:** Compact and intricate designs like the jug-shaped radiating element may require precise manufacturing techniques, increasing production costs.

4.6 Applications

1. **Ultra-Wideband Wireless Communication Systems:** Suitable for high-speed data transmission and low-latency applications.
2. **Portable Devices:** Ideal for integration into smartphones, laptops, and IoT devices due to its compact size and efficiency.
3. **Radar and Imaging Systems:** Can be used in short-range radar applications and imaging systems due to its wide frequency range and minimal distortion.
4. **Medical Devices:** Applicable in medical telemetry and imaging due to its low power and compact nature.

CONCLUSION

This project presents the design, analysis, and evaluation of a miniaturized coplanar waveguide (CPW)-fed ultra-wideband (UWB) Jug-Shaped Antenna tailored for wireless applications. It addresses the challenges of miniaturization, wide bandwidth, and efficient performance. The antenna is fabricated on an FR4 substrate with a dielectric constant of 4.4, making it ideal for modern wireless systems with limited space. Simulation and measurement results reveal excellent impedance matching and return loss characteristics. The CPW-fed UWB antenna demonstrates excellent return loss performance across the frequency bands 2.6383–3.6896 GHz, 4.6596–5.0724 GHz, and 6.5805–9.4254 GHz, ensuring efficient impedance matching and minimal power reflection. This performance highlights the antenna's capability to support a wide range of applications, including Wi-Fi, 5G communication, satellite communication, radar systems, and remote sensing. The CPW-fed design further enhances integration into compact and modern systems. Overall, the antenna's low return loss and versatility make it a reliable choice for multi-band and ultra-wideband applications. The Jug-Shaped Antenna combines miniaturization, wideband operation, and efficient performance, making it a strong candidate for next-generation wireless communication systems.

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